

```
WHERE username = NEW.username AND mailbox = NEW.mailbox;
IF FOUND THEN
RETURN NULL;
ELSE
RETURN NEW;
END IF;
END;
$function$
```

```
CREATE OR REPLACE TRIGGER mergeexpires BEFORE INSERT ON
expires FOR EACH ROW EXECUTE PROCEDURE merge_expires();
```

The above trigger code was taken from Dovecot Wiki <http://wiki2.dovecot.org/Plugins/Expire>

Expires table is used for deleting old mail from mailboxes. Particularly Trash and Spam, like every 30 days or so.

Function: *sender_bcc_map*

This function is optional. In my set-up, Postfix is configured to send a BCC to <sender>+Sent@<domain> for every mail sent as an authenticated user. This later gets classified as 'Sent' mail in Dovecot. If you don't want the server generating a sent-mail copy automatically, you don't need this.

```
CREATE OR REPLACE FUNCTION sender_bcc_map(sender character
varying)
RETURNS SETOF character varying
LANGUAGE plpgsql
AS $function$declare
username text;
domainname text;
dom_id domains.id%type := 0;
s_username text;
begin

username := split_part(sender, '@', 1);
domainname := split_part(sender, '@', 2);

perform 1 from users where name = username and domain_id =
(select id from domains where name = domainname);

if found then
s_username := username || '+Sent';
return next s_username || '@' || domainname;
end if;

select destination || '+Sent' into s_username from user_
aliases where name = username and domain_id = (select id from
domains where name = domainname);

if found then
return next s_username || '@' || domainname;
end if;
```

```
return;

end;$function$
```

Function: *sasl_login*

This function is used by Postfix to check whether a sender is authorised to send the message at the address specified by the user. The function returns a list of addresses that the particular authorised user may send:

```
CREATE OR REPLACE FUNCTION vmail.sasl_login(sender character
varying)
RETURNS SETOF character varying
LANGUAGE plpgsql
AS $function$declare
s_username vmail.users.name%type;
s_domainname vmail.domains.name%type;
begin

perform 1 from vmail.active_users_userdb t where t.user =
sender;

if found then
return next sender;
end if;

select ua.destination, d.name into s_username, s_domainname
from vmail.user_aliases ua join vmail.domains d
on ua.domain_id = d.id
where ua.name = split_part(sender, '@', 1) and
d.name = split_part(sender, '@', 2);

if found then
return next s_username || '@' || s_domainname;
end if;

return;
end;$function$
```

We're now done with the PostgreSQL setup. That is a lot of SQL code. But, there is still something missing—we need to delete aliases to a user when a user is deleted, as well as check for an existing alias when a new user is created. Both of these are left for readers to try out as an exercise. :-)

In the next part, we'll look at how to configure the other aspects of a mail system starting with Dovecot. **END** 

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